

1. Overview

This assignment involved cleaning raw transaction data, computing multiple revenue layers, analyzing settlement exposure, and building an interactive dashboard using Power BI.

The objective was to transform raw payment data into meaningful business insights while maintaining financial accuracy and dynamic modeling.

The solution was built in a structured and modular way to ensure clarity, scalability, and proper financial logic implementation.

2. Data Cleaning & Preparation

The dataset provided was raw and required cleaning before analysis.

The following steps were performed:

- Standardized column names
- Corrected data types (Date, Decimal, Text)
- Handled null and missing values
- Created a safe date column to avoid parsing errors
- Normalized pricing fields

Pricing was provided in both Flat and Percentage formats.

To standardize calculations, all pricing values were converted into absolute INR per transaction:

- If pricing type = Flat → used directly
- If pricing type = Percentage → $\text{Transaction Amount} \times \text{Rate} \div 100$

This ensured consistent revenue computation.

3. Revenue Logic Implementation

Revenue layers were computed as per the business definitions:

- $\text{ERP Revenue} = \text{Merchant Pricing} - \text{Partner Pricing}$
- $\text{Edviron Net Revenue} = \text{Partner Pricing} - \text{Edviron Buying}$
- $\text{Edviron Gross Revenue} = \text{ERP Revenue} + \text{Edviron Net Revenue}$

Each revenue was first calculated at the transaction level and then aggregated using DAX measures.

Settlement-related calculations were also implemented:

- ERP Payable
- Edviron Retained
- Pending Exposure
- Capture Pending logic

Where assumptions were required, they were clearly defined and applied consistently.

4. Reports & Business Insights

Multiple analytical views were built:

- Daily / Monthly Revenue Summary
- Partner-wise performance
- Gateway-wise contribution
- Payment method mix
- Pending vs Settled exposure

These reports provide insight into transaction volume, revenue distribution, margin behavior, and operational risk exposure.

5. Interactive Dashboard

A final Executive Dashboard page was created containing:

KPI Cards:

- Total Transactions
- Total GMV
- ERP Revenue
- Edviron Net Revenue
- Edviron Gross Revenue
- Pending Exposure
- Unique Users
- Payment Frequency

Visuals:

- Revenue Split (ERP vs Edviron)
- Time-based Revenue Trend (Year → Month → Day drill)
- Gateway Contribution
- Payment Method Mix

Filters:

- Date Range Selector
- Partner Filter
- Gateway Filter
- Payment Method Filter

All visuals and KPIs dynamically update based on slicer selections.

6. Modeling & Technical Approach (Part 5)

All key metrics were implemented using DAX measures rather than static columns. This ensures automatic recalculation based on filter context.

Example:

Total GMV = SUM(Final_Transactions[Transaction Amt])

The model was designed in a modular structure:

- Pricing normalization layer
- Revenue calculation layer
- Settlement logic layer
- Aggregated KPI measures

No hard-coded values were used in calculations. All metrics derive directly from transactional and pricing data.

The dashboard updates automatically when filters change and recalculates correctly when the dataset is refreshed.

7. Assumptions

1. Percentage pricing is applied on transaction amount.
2. Gateway settles full amount to Edviron before ERP settlement.
3. ERP payable is derived from calculated ERP revenue unless explicitly stated.
4. Negative revenue values reflect real margin scenarios.

8. Conclusion

The final solution is dynamic, financially accurate, modular, and user-friendly.

It provides both operational insight and executive-level visibility into revenue performance and settlement exposure.

The dashboard is fully interactive and designed for business usability.